

Call for Workshop Proposals – IEEE BIBM 2026

AutoML for Medical Data Analysis and Healthcare Intelligence

Workshop Description

Automated Machine Learning (AutoML) has emerged as an important paradigm for building intelligent systems with minimal human intervention. By automating critical stages of the machine learning pipeline, including feature engineering, model selection, hyperparameter optimization, and neural architecture search, AutoML provides efficient and scalable solutions for complex biomedical and healthcare problems. With the rapid growth of healthcare and biomedical data, such as electronic health records, medical imaging, omics data, and physiological signals, there is an increasing demand for automated methods capable of handling high dimensionality, heterogeneity, noisy data, and limited annotations. This workshop aims to bring together researchers and practitioners from machine learning, biomedical informatics, and healthcare analytics to discuss recent progress, key challenges, and future directions of AutoML for biomedical data analysis.

Topics of Interest

- AutoML for biomedical informatics and healthcare analytics
- Automated learning for medical imaging and bioinformatics
- Neural architecture search in biomedical applications
- AutoML for multi-modal and heterogeneous biomedical data
- Trustworthy and interpretable AutoML in healthcare systems
- AutoML for electronic health records and clinical data mining
- Efficient and scalable AutoML for biomedical big data
- Learning under noisy, sparse, or uncertain biomedical labels
- Meta-learning and transfer learning for biomedical AutoML
- AI-driven clinical decision support using automated machine learning

Workshop Format

Full online workshop including keynote talks, paper presentations, and interactive discussions.

Workshop Organizers

Name	Affiliation	Email
Nan Li	Shanxi University, China	linan10@sxu.edu.cn
Xinyan Liang	Shanxi University, China	liangxinyan@sxu.edu.cn
Jiang Huamao	The First Affiliated Hospital of Jinzhou Medical University, China	jiahuamao@jzmu.edu.cn
Wellington P. Santos	Universidade Federal de Pernambuco, Brazil	wellington.santos@ufpe.br

Program Committee

- **Huixiang Zhen** — China University of Geosciences (Wuhan), China
- **Zhicheng Wu** — Victoria University of Wellington, New Zealand
- **Tong Guo** — Nanyang Technological University, Singapore
- **Jiading Wang** — Fujitsu Limited, Japan
- **Aohan Mei** — Northeastern University, China

Important Date

Workshop Proposal Submission Deadline: June 25, 2026

Contact Information

linan10@sxu.edu.cn

liangxinyan@sxu.edu.cn

Jianghm@jzmu.edu.cn

wellington.santos@ufpe.br